INCH-POUND

MIL-P-70997 (AR)
4 May 1991

MILITARY SPECIFICATION

PROJECTILE, 81MM, HE, M821E1/M889E1 BODY ASSEMBLY

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for the projectile, M821E1 or M889E1, for use in the 81MM Mortar.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-S-50783 - Steel Alloy, Special Purpose for Ammunition Components (HF-1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL

TT-C-490 - Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings

STANDARDS

MILITARY

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components
MIL-STD-1459 - Macrograph Standards for Steel Bars, Billets, and Blooms for Ammunition Components
MIL-STD-1949 - Inspection, Magnetic Particle

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.6)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12630554 Projectile, HE, M821E1 or M889E1, Body Assembly

(Copies of drawings, publications, and other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-government documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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PUBLICATIONS

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E-8 - Tension Testing of Metallic Materials
A-322 - Hot Rolled Alloy Steel Bars
A-350 - Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron and Wrought Iron
B-117 - Standard Test Method of Salt Spray (Fog) Testing
E-381 - Macroetching Metals and Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

REQUIREMENTS

3.1 Materials. Materials, shall be in accordance with the applicable specifications drawings and this specification. Bar stock shall be free of cracks splits and other similar defects. For HF 1 the internal soundness of the steel supplied shall comply with MIL-S-50783, except that macroetch quality shall be equal or better than standard D-2 of MIL-STD-1459 only; all others are unacceptable. Each billet shall be macroetched in accordance with ASTM-E381.

3.2 Parts and assemblies. Parts and the projectile assembly shall comply with all requirements specified on drawing (dwg. 12630554) and associated drawings, and with all requirements specified in applicable specifications and standards. Threads and other surfaces shall be free of burrs or foreign matter which may interfere with further assembly or proper functioning.

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3.3 Mechanical properties. All components shall be uniformly heat treated to meet the mechanical properties requirements specified on drawing 12630579. Hardness testing shall be used as an indicator of product uniformity.

3.4 Hydrostatic test. Each projectile body shall be able to withstand the hydrostatic load specified on dwg. 12630579 without cracking, leakage, rupture or permanent deformation when tested in accordance with 4.4.3.5 and 4.5.4.

3.5 Projectile defects.

3.5.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, solid shuts, pipes, porosity, inclusions, tears, blowholes, folds, seams and other similar defects when inspected by magnetic particle inspection or "white" light in accordance with 4.4.3.6 and 4.5.5.1.

3.5.1.1 Body cavity. In addition to the requirements of 3.5.1, the body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, sharp edges and other similar defects.

3.5.2 Visual standards (internal and external). Visual standards for evaluation of metal defects after painting shall be established within 90 days after the start of production. However, the requirements for compliance with 3.5.1 and 3.5.1.1 are independent of visual standards.

3.6 Proving ground (maximum average pressure) test. Projectile body assemblies shall reveal no evidence of failure when tested at maximum average pressure in accordance with 4.5.7. Body assembly failure shall be:

a. Projectile body deformation attributable to noncompliance with drawing or specification requirements.

b. Loss or breakup of any component, in the mortar tube or in flight.

c. Short round due to undetermined cause and projectile not recovered.

3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (see 4.3 and 6.2).

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3.8 Workmanship. All parts and assemblies shall be fabricated in a thorough workmanlike manner. Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings. No part shall have a burr which might be injurious to personnel handling the item. No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, corrosion or other foreign matter remains therein after final cleaning. Projectile assemblies shall be prepared for painting in accordance with TT-C-490. The cleaning method used shall not be injurious to any part nor shall parts be contaminated by the cleaning agent. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification. After painting, the body shall be free from bare spots, blisters, grease, dirt and other foreign matter. There shall be no pool of paint in the bottom of the cavity. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for

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acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I. Total quantities are as follows:

Part description	Drawings	Quantity
Body, prior to machining obturating band groove	12630579	10
Body	12630579	10
Projectile, 81mm, HE, M821E1 body assembly	12630554	30
Tensile specimen	See ASTM E-8	8
Macroetch specimen	- -	4

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I, First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Bar Stock			
	Macroetch Test	4	3.1	Visual/4.4.3.1
	Chemical Composition	See Note 1	3.1	Gage/4.4.3.2
	Body, Prior to Machining Obturating Band Groove (dwg. 12630579)			
	Examination for defects	10	3.2	4.4.2.2
	Hydrostatic Test	10	3.4	4.4.3.5/4.5.4
	Tension Test (specimen quantity)			
	Hardest body	4	3.3	4.4.3.3/4.5.2
	Softest body	4	3.3	4.4.3.3/4.5.2
	Body (dwg. 12630579)			
	Examination for Defects	10	3.2	4.4.2.3
	Projectile Metal Defects (magnetic particle)	10	3.5	4.4.3.6/4.5.5.1
	Projectile, XM821E1, After Painting (dwg. 12630554)			
	Examination for Defects	25	3.2	4.4.2.4
	Salt Spray Test	3	3.2	4.4.3.7/4.5.6
	Paint adhesion test	2	3.2	4.4.3.7
	Maximum Average Pressure Test (proving ground)	25	3.6	4.4.3.8/4.5.7

NOTES: 1/ Four from one mill heat of steel. Two shall be done by mill in accordance with MIL-S-50783 "check analysis" and two selected randomly by the contractor shall be analyzed by an independent lab.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall conform with the lot formation provisions of MIL-A-48078 and the following:

4.4.1.1 Heat treatment batches. Heat treatment batches shall contain bodies from not more than one mill heat of steel, that have been heat treated in the same heat treatment equipment, at the same temperature conditions in one unchanged process. No projectile shall be re-heat treated if it fails to meet the applicable mechanical properties requirements without government approval. The batch thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Projectile lot. All heat treatment batches that have complied with the mechanical properties requirements of drawing 12630579 may be re-grouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing revision and same specification revision. A projectile lot may contain projectiles from more than one mill heat of steel.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes Sampling Inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs:

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TABLE II. 0-1 Attributes Plans

LOT SIZE	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Tool control inspection. Dimensions marked tool control (*) on dwg. 12630579 shall be gaged during First Article inspection, at the beginning of each workday, every four (4) hours during production, for the last component prior to and first component after each tooling, equipment, or process change, repair or adjustment. Measurements of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimension prior to substitution. When acceptability has been determined, inspection of tool dimensions will be permitted. Whenever a defective tool control dimension is found, immediate action will be taken by the contractor to correct the tool problem and all material produced since the last successful inspection shall be rejected.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Bar Stock	1 OF 1	- -
			NEXT HIGHER ASSEMBLY: 12630579

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Chemical composition incorrect	4 (0-1)	3.1	Gage/4.4.3.2
B	Bar stock cracked on outside diameter	100%	3.1	Visual
C	Insufficient cropping of billet ends	100%	3.1	Visual/4.4.3.1
Major	None defined			
Minor	None defined			

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Prior to Machining Obturator Groove	1 OF 2	12630579
			NEXT HIGHER ASSEMBLY 12630554

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Hydrotest	100%	3.4	Gage/4.4.3.5
Major				
101	Length of body from front face to rear shoulder	III	3.2	Gage
102	Major diameter of rear thread undersize	100%	3.2	Gage
103	Pitch diameter of rear thread	III	3.2	Gage
104	Outside diameter of shank at rear of body	III	3.2	Gage
105	Runout of aft thread with bourrelet and rear shoulder at two lengths	III	3.2	Gage
106	Runout of fwd thread with bourrelet and front face at two lengths	III	3.2	Gage
107	Minor diameter of fwd thread	III	3.2	Gage
108	Pitch diameter of fwd thread	III	3.2	Gage
119	Outside diameter at basic lengths	III	3.2	Gage
110	Length of thread boss	III	3.2	Gage
111	Diameter of bourrelet	III	3.2	Gage
112	Tension test	8 (0-1)	3.3	Gage/4.4.3.3
113	Hardness test	100%	3.3	Gage/4.4.3.4
114	Wall thickness at five basic lengths	III	3.2	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Prior to Machining Obturator Groove	2 OF 2	12630579
			NEXT HIGHER ASSEMBLY 12630554

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
115	Wall thickness min., near base of cavity	III	3.2	Gage
116	Distance from base of body cavity to end of thread boss	III	3.2	Gage
117	Runout of rear taper with bourrelet at basic length	III	3.2	Gage
118	Runout of ogive with bourrelet at basic length	III	3.2	Gage
119	Forward thread not full for required length	III	3.2	Gage
120	Rear thread effective length, min.	III	3.2	Gage 1/
Minor				
201	Lead in angle of forward thread	V	3.2	Gage
202	Radii or chamfers missing or improper	V	3.2	Visual/6.5
203	Depth of centering hole on thread boss, max.	V	3.2	Gage
204	Angle of drill point on thread boss, max	V	3.2	Gage
205	Surface finish improper	V	3.2	Visual/6.5
206	Evidence of poor workmanship	V	3.8	Visual

NOTES: 1. Major diameter shall be checked to full form for min. length.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Body, After Machining Obturator Groove	1 OF 1	12630579
			NEXT HIGHER ASSEMBLY 12630554

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None Defined			
Special				
A	Metal defective (magnetic particle inspection)	1/	3.5	Visual/4.5.5.1
Major				
101	Length of flat, min., forward of obturator groove	III	3.2	Gage
102	Width of obturator groove at basic depth	III	3.2	Gage
103	Outside diameter within obturator groove	III	3.2	Gage
104	Runout of obturator groove with bourrelet surface	III	3.2	Gage
105	Lead-in angle of obturator groove	III	3.2	Gage
106	Outside diameter rear of obturator groove not flush or below bourrelet diameter	III	3.2	Gage
107	Length from front face to obturator groove	III	3.2	Gage
Minor				
201	Radii missing or improper	V	3.2	Visual/6.5
202	Surface finish improper	V	3.2	Visual/6.5
203	Evidence of poor workmanship	V	3.8	Visual

NOTES: 1. See 4.4.3.6

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.4	TITLE Projectile, HE, M821E1 or M889E1 Body Assembly	SHEET 1 OF 1	DRAWING NUMBER 12630554
			NEXT HIGHER ASSEMBLY 12630553

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Metal defective (white light inspection)	100%	3.5	Visual/4.5.5.2
B	Absence of hydrotest stamp	100%	3.2	Visual
Major				
101	Outside diameter, max.	III	3.2	Gage
102	Presence of protective finish on threads	III	3.2	Visual
103	Presence of burrs in forward or rear threads	III	3.2	Visual
104	Corrosion preventative compound missing from threads, front face or rear shoulder	III	3.2	Visual
105	Protective finish missing or inadequate on or in projectile	III	3.2	Visual
106	Failure of protective finish (salt spray and adhesion tests)	1/	3.2	Visual/4.5.6
107	Proving ground test (maximum average pressure)	2/	3.2	Gage/4.5.7
Minor				
201	Presence of dents	V	3.2	Visual
202	Evidence of poor workmanship	V	3.8	Visual

NOTES: 1. See 4.4.3.7.
2. See 4.4.3.8.

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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4.4.3 Testing. The following tests shall be performed in accordance with the provisions of Table I and section 4.4.

4.4.3.1 Macroetch. A full cross section shall be cut from both ends of each billet for comparison with the standards prescribed in 3.1. Test methods and procedures shall be as prescribed in ASTM E-381. Failure of any specimen shall be cause for removal of more material from the billet until the requirements in 3.1 are met.

4.4.3.2 Chemical Composition. After delivery of material the contractor (not mill) shall ramdomly select two specimens per mill heat of steel and forward them to an independent laboratory to verify mill certification of compliance with the composition requirements contained in MIL-S-50783. Actual composition results (not only certification) shall be made available to the government representative. Test methods and procedures shall be in accordance with the test procedures specified in ASTM-A322 referencing ASTM-E350. The mill shall select two specimens in accordance with MIL-S-50783.

4.4.3.3 Tension test. (See 4.5.2) At startup at least fifty (50) projectiles shall be hardness tested in accordance with 4.4.3.4. From the hardest and softest projectiles, four tensile specimens shall be manufactured from each body, eight total. When it has been determined that the applicable requirements for tensile strength and elongation as specified on the drawing have been met, the acceptable range for hardness shall be considered established for projectile batches from one mill heat of steel produced in accordance with 4.4.1.1. Should projectiles be found subsequently with a higher or lower hardness than in the established range, then a new set of tension test specimens must be prepared and tested in order to extend the acceptable hardness range. If heat treatment or stress relief conditions have varied from designated process limits, projectiles from the "non-conforming" process batch shall be segregated from batches from the same mill heat which conformed to nominal process limits. New tensile specimens shall be made from the hardest and softest of the first fifty projectiles (minimum) in the non-conforming batch. Projectiles shall be accepted from that batch if their hardness falls within the hardness range of the hardest and softest "non-conforming" projectiles which passed the tensile test. Once "nonconforming" bodies are qualified, they may be included in a projectile lot (see 4.4.1.2).

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4.4.3.4 Hardness test. (see 4.5.3) Each projectile shall be hardness tested to confirm uniformity of the heat treatment/ stress relief operations and for the purpose of selecting the projectiles used for the tensile test. Projectiles will be tested two times (minimum) in the zone where the obturating band groove will later be machined.

4.4.3.5 Hydrostatic test. (see 4.5.4) Prior to machining the obturating band groove but after projectiles are fully machined to the extent that wall thickness is at the required final size in all other locations, each projectile shall be hydrostatically tested to determine compliance with the requirements of 3.4.

4.4.3.6 Magnetic particle inspection. (see 4.5.5.1) After stress relief, hydrotesting, and machining the obturating band groove, bodies shall be inspected by magnetic particle inspection techniques approved by the responsible technical agency according to the following sequence:

a. 100 percent magnetic particle inspection will be in effect until 2500 consecutive bodies have been inspected and found to be free of metal defects.

b. Reduced inspection may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten connsecutive working days.

c. If no defect is found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive working days.

d. If no defect is encountered during the five days, the sample may be reduced to 5 bodies per hour.

e. If a defect is encountered by magnetic particle inspection during reduced sampling (b through d) the magnetic particle inspection shall revert back to (a) above. In addition a 100 percent back screening to the last acceptable hours production will be performed.

f. For requalification of reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a through d) above.

g. Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible agency.

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h. Continual application of a 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as an alternative procedure in lieu of corrective material or manufacturing process action to eliminate a continuing material or process induced defect. If corrective action is not instituted and the cause eliminated, the government reserves the right to withdraw acceptance inspection until the contractor can prove his materials and process are capable of producing suspect-free projectiles.

4.4.3.7 Salt spray test. (See 4.5.6) In performing the following tests, TT-C-490 shall apply unless specified otherwise herein. The test shall be performed for first article inspection, each day of painting during production and whenever a change is made to the paint process, until all samples from five consecutive days of painting have passed the test. Thereafter the sample frequency may be reduced to twice per week during continuous days of painting. Five specimens shall be selected, consisting of randomly chosen projectiles which have been process through all the same cleaning, phosphating, painting and drying steps at the same time using the same equipment as other production projectiles. The samples shall be taken from each continuous final paint operation up to 24 hours long. One projectile shall be taken during the first thirty (30) minutes of painting, one during the last thirty (30) minutes and the others at any other time.

Prior to salt spray testing, two (2) of the specimens must only pass an adhesion test using the procedures and pass-fail criteria contained in 4.2.8 and 3.5.6 respectively, of TT-C-490. If both specimens pass the adhesion test, the three (3) remaining specimens shall be scored lengthwise using a precision scribing tool/insert or equivalent in preparation for the salt spray test.

4.4.3.8 Proving ground (maximum average pressure) test. See 4.5.7 Twenty-five (25) projectiles shall be tested from the First Article sample and ten (10) from each of the first three (3) production lots and from three (3) consecutive lots when a new mill heat of steel is introduced into production. However, any lot may be subjected to this test, retested, or testing may be reinstituted when so directed by the procuring contracting officer.

An attempt will be made to recover at least 60 percent of the projectiles fired in each test. If a round falls short due to an undetermined cause and is not recovered it shall be classed defective. Defect classification and individual accept/reject criteria shall be in accordance with Table III.

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TABLE III Proving Ground Test

Category	Defect	Accept/Reject	
Critical	Premature burst, metal part breakup or deformation due to launch forces	0	1
	Horizontal range less than 80% of average range	0	1

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection table herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Tension test. (See 4.4.3.3) Four (4) specimens shall be produced from each projectile selected from the tension test. Specimens shall be drawn from opposite side of the body from both the boattail and ogive zones and shall be the largest obtainable flat or round standard or subsize specimen conforming to ASTM-E8. Equipment and test procedures shall be as prescribed by ASTM-E8 except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0). Failure of any specimen to achieve the required tensile strength or elongation shall be cause for rejection of all material from the same mill heat of steel. Actual test data from an independent source (not only mill certification) must be made available to the government representative.

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4.5.3 Hardness Test. (See 4.4.3.4) The test shall be performed using any automated hardness tester which will produce accurate repeatable results with readings for an acceptable projectile between 25-80% of the full scale capability of the test equipment. Operating procedures shall be as specified by the applicable equipment manual, standard ASTM test methods and the following subject to approval by the government representative:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a depth sufficient to remove the decarburized zone. (Note: Impressions shall be completely removed by machining of the obturating band groove on final machining.)

b. The hardness tester calibration shall be tested at start up, at least once each four hours of continuous operation, and after a layoff of three or more hours.

c. The body shall be fixtured so that no rocking, shifting or deflection of the body will occur during the test.

d. The hardness tester shall not produce indentations deeper than 25% of the indentor ball diameter or wall thickness of the projectile in that zone.

e. Hardness readings shall be taken at least ten indentor ball diameters apart. At least two readings shall be taken per projectile and averaged.

4.5.4 Hydrostatic test. (See 4.4.3.5) Each projectile body shall be tested using equipment capable of performing the test accurately and consistently. During the test, the pressure will be applied gradually until the required minimum pressure is reached and held for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continous operation. It shall also be calibrated prior to restarting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower that required by the applicable test, the bodies tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Projectile rupture, cracking, leakage or bulging during the test shall be cause for rejection of the body. The test fixture shall not restrict the projectile body from expanding longitudinally or radially under pressure. Protective barriers should be installed to shield operators from metal fragments which may come from defective projectiles that fail the test.

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4.5.5 Metal defects inspection.

4.5.5.1 Before paint (Magnetic particle inspection). (See 4.4.3.6) Projectiles shall be tested using procedures for the wet fluorescent method as specified in MIL-STD-1949. Each projectile shall be tested twice. Both transverse and longitudinal fields shall be induced in the projectile using any method to establish the fields. Between the two (2) tests and following the second test, each projectile shall be demagnetized. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 militers (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 mls of solids per 100 mls suspension when water is used. After draining the excess suspension, the body shall be inspected under ultraviolet light for the presence of defects. The requirement of MIL-STD-1949 to check system effectiveness by means of a steel ring specimen shall be waived. Components which exhibit indications which are suspected of being shallow surface defects may have material removed to a maximum of 0.005 inches depth subject to the following restrictions:

a. Involved areas shall be blended with adjacent surfaces to insure that no sharp edges, steps or gouges remain.

b. The involved areas shall be independently checked for verification that all drawing dimensions (wall thickness, variations, etc.) are within acceptable limits.

c. The components shall be again subjected to magnetic particle inspection for verification of defect removal.

4.5.5.1.1 Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level III.

4.5.5.2 After paint ("white" light). Each projectile shall be visually inspected internally and externally for the defects described in 3.5 using any type of lighting which will facilitate good inspection of all surfaces. If any projectile defects are found during this inspection, the magnetic particle sampling in 4.4.3.6 shall revert back to section "a".

4.5.6 Salt spray test. (See 4.4.3.7) The salt solution, test equipment and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise herein. Specimens shall be positioned between 15 and 30 degrees from vertical during the test. No coating of any kind shall be

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permitted over the paint on stamp marks of the specimens prior to testing, however areas in contact with racks or supports, unpainted surfaces and areas within 3/16 inch of unpainted surfaces may be protected.

When examining the specimens for signs of paint failure, the entire surface of the projectile shall be considered the significant surface.

As used herein, the term "blister" refers to an unbroken area of paint with a bubble-like appearance whereas "pits" or "spots" are exposed phosphate pretreatment or bare metal visible through holes in the paint. Blisters which are present when the test is over are to be counted within five (5) minutes of removal from the test chamber.

Projectiles shall be tested for the duration specified on drawing 12630554. After testing, any projectile which fails for any of the following reasons shall cause rejection of all parts painted since the last successful test and a return to the original sampling plan if failure occurred during reduced sampling. The failure criteria in a-i applies to the significant surfaces only.

a. The paint shows more than 1/8 inch creepage, blistering or loss of adhesion of paint from the scribe mark, stamp mark, or edge.

b. There are more than eight (8) blisters, total.

c. Any blister exceeds 1/16 inch average diameter.

d. Any pit exceeds 1/8 inch average diameter.

e. There are four (4) or more scattered pits exceeding 1/16 inch average diameter.

f. There are four (4) or more spots within one square inch.

g. There are three (3) spots within one square inch in two (2) or more places.

h. There are more than six (6) scattered spots, total.

i. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 1/4 inch in diameter total area.

j. There is some variation of the above or an unlisted defect anywhere which the government representative deems excessive.

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4.5.7 Proving ground (maximum average pressure) test. (see 4.4.3.8) The samples supplied in accordance with Table I and 4.4.2.4 (inert loaded to the service weight and dummy fuzed) shall be assembled into complete rounds and fired for recovery from a mortar for which the round is standard, the tube of which has a minimum of 25 percent life remaining based on current mortar tube condemnation criteria and propelling charges adjusted to produce 13,189-14,157 psi at 70°F. Photographic equipment shall be set up near the muzzle to record the flight of the projectile to help determine compliance with 3.6. The test shall be performed at a government proving ground in accordance with applicable USA TECOM acceptance test procedures. . Cartridges shall be conditioned at 70 degrees plus or minus 5 degrees Fahrenheit for 24 hours prior to firing.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of level C of MIL-STD-1169.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use with the 81mm mortar on the cartridges, 81mm, HE, M821E1 and M889E1.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of test data. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, U.S. Army Armament, Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAT-B (D) and SMCAR-CCH-P, Picatinny Arsenal, NJ 07806-5000.

6.4 Submission of inspection equipment designs for approval. See paragraph 6.2.3 of MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament, Munitions and Chemical Command (AMCCOM), ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability. The roughness comparison specimens prescribed by ANSI-B46.1 may be used as a basis for surface roughness determination when applicable.

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6.6 Drawings. Drawings listed in section 2 of this specification under the headings U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Proving ground test summary.

Test	Sample Size	Requirements
First Article/ Max. Average Pressure	25 Inert M821E1 Cartridges with Dummy Fuze, Acceptable Ignition Cartridges and 100 Modified M220 Propelling Charges. (Warmer rounds as required)	See 3.6

Test	Sample Size	Requirements
Production Lots/ Max. Average Pressure	10 Inert M821E1 Cartridges with Dummy Fuze, Acceptable Ignition Cartridges and 40 Modified M220 Propelling Charges. (Warmer rounds as required)	See 3.6

6.8 Subject term (key word) listing.

Ammunition, mortar
Batch, heat treatment
Body
Groove, obturating band
Heat of steel, mill
HF-1
Indications
Pressure, maximum average
Projectile
Steel
Zone, decarburized

Custodian:
Army-AR

Preparing activity:
Army-AR

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70997 (AR)	4 May 1991

3. DOCUMENT TITLE
PROJECTILE , 81MM, HE, M821E1/M889E1 BODY ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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